

Parasitism Rates of Spruce Budworm Larvae: Testing the Enemy Hypothesis Along a Gradient of Forest Diversity Measured at Different Spatial Scales

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Abstract

Vegetational diversity is generally thought to be associated with ecosystem stability and resilience to perturbations such as insect outbreaks. The *enemies' hypothesis* states that vegetational diversity contributes to greater top-down control of insect pests, by providing further resources to natural enemies than homogeneous environments. However, direct evaluation of this hypothesis is difficult because different species of natural enemies can respond to vegetational diversity in dissimilar manners and at different spatial scales depending on functional traits such as prey/host specificity and dispersal. In this study, we specifically test the *enemies' hypothesis* at the landscape level in a continuous forest environment. We investigated how parasitism of spruce budworm larvae by the common parasitoids *Apanteles fumiferanae* and *Glypta fumiferanae* vary with forest diversity and host larval density at different spatial scales in the province of Quebec (Canada). We found that parasitism rates of the two parasitoid species we examined respond in opposite ways to forest diversity. Parasitism by *A. fumiferanae* was positively related to forest diversity, whereas parasitism by *G. fumiferanae* was negatively related to forest diversity. In agreement with the *enemies' hypothesis*, we also found that spruce budworm larval density decreased with forest diversity. We discuss these results with respect to the *enemies' hypothesis* and the presumed host range of the parasitoids species we examined, as well as their body size. Because *A. fumiferanae* kills its host earlier than *G. fumiferanae*, we conclude that northern forest landscapes could be more affected by spruce budworm defoliation than southern forests during the present and future outbreaks.

Key words: *enemies' hypothesis*, vegetational diversity, parasitoid, spruce budworm, spatial scale

Biological diversity is generally thought to be associated with ecological mechanisms that promote ecosystem stability and resilience to perturbations (McCann 2000, Loreau and de Mazancourt 2013, Jactel et al. 2017). For example, outbreaks of forest insect pests are more common in species-poor boreal forests versus species-rich tropical forests (Odum 1953, Elton 1958). The *enemies' hypothesis* (Root 1973) suggests that vegetational diversity is positively related to the abundance of generalist (i.e., polyphagous) natural enemies by providing a greater abundance and diversity of alternative prey species, microhabitats, and other resources such as nectar (Russell 1989). As a result, vegetational diversity can contribute indirectly to greater top-down control of insect pests.

Outbreaks of plant-feeding insects are important large-scale natural disturbances affecting boreal and sub-boreal forest ecosystems throughout the Northern Hemisphere (Cooke et al. 2007, Netherer and Schopf 2010). As such, they are major drivers of forest succession (Baskerville 1975, Bouchard et al. 2006), ecosystem service provisioning (Maguire et al. 2015), food-web structure (Eveleigh et al.

2007), and carbon dynamics (Kurz et al. 2008, Dymond et al. 2010). However, in some contexts, it becomes important to intervene to reduce outbreaks' intensity or duration to reduce wood losses and associated economic impacts. Although increasing vegetational diversity has mostly been used to manage insect pests in agroecosystems (Russell 1989, Bianchi et al. 2006, Chaplin-Kramer et al. 2011), the strategy has also been suggested to control pests affecting forests (Jactel and Brockerhoff 2007).

Assessing the relationships between vegetational diversity and top-down control of insect pests is not straightforward. One associated challenge is that specialist (i.e., oligophagous) natural enemies may be more effective than generalists in controlling herbivore populations. As a result, it is important to determine the relative responses to specialist and generalist natural enemies to vegetational diversity (Sheehan 1986). Another challenge is that the ecological mechanisms that underlie the *enemies' hypothesis* include complex trophic interactions whose dynamics vary with spatial context (Holt 2002, Bommarco and Banks 2003, Cronin and Reeve 2005,

McCann et al. 2005). Further, different species of natural enemies perceive their habitat at different spatial scales (Cronin and Reeve 2005). Larger enemies tend to have greater dispersal abilities and are able to perceive landscape structure at larger spatial scales (Roland and Taylor 1997). Similarly, specialists tend to respond at smaller spatial scales than generalists (Chaplin-Kramer et al. 2011). Adopting a species-specific and multi-scale perspective thus allows to evaluate how vegetational diversity affect each natural enemies' efficiency (e.g., attack rates), as well as the spatial scales at which these effects are found (Meynard et al. 2013).

In eastern North America, the native eastern spruce budworm (*Choristoneura fumiferanae* (Clemens); Lepidoptera: Tortricidae) is the most important defoliator of fir and spruce forests in the boreal and sub-boreal regions (Rowe 1972, Bouchard and Auger 2014). Several consecutive years of severe defoliation result in widespread tree damage and mortality (MacLean and Ostaff 1989, Gray 2008, Bouchard and Auger 2014), with important economic consequences for the forest industry (Chang et al. 2012). Outbreaks are cyclic, with intervals of 30–40 yr between peaks (Blais 1983, Jardon et al. 2003, Boulanger and Arseneault 2004). Trophic interactions between the spruce budworm and its natural enemies—mainly insect parasitoids—play a significant role in population cycling (Royama et al. 2017), yet many aspects remain poorly understood.

Forest vegetational diversity has been shown to be associated with slower expansion rates of spruce budworm outbreaks (Bouchard and Auger 2014), lower outbreak intensity (Robert et al. 2018), and reduced probability of tree mortality (Su et al. 1996, Cappuccino et al. 1998). This effect of forest diversity on outbreaks can be in part explained by the *enemies' hypothesis*, which states that the abundance and efficacy of natural enemies vary as a function of vegetational diversity. For example, Quayle et al. (2003) showed that parasitism of spruce budworm eggs by *Trichogramma minutum* Riley (Hymenoptera: Trichogrammatidae), a polyphagous species (Laing and Corrigan 1995), was positively related to the proportion of nonhost basal area measured at the stand level. Similarly, Cappuccino et al. (1998) found that parasitism rates of late larval and pupal spruce budworm were higher in coniferous stands surrounded by deciduous forests than in extensive coniferous landscapes.

Forest diversity also affects the rate of change in spruce budworm parasitoid community structure during the release phase of outbreaks (Eveleigh et al. 2007). As spruce budworm densities rise, the number of generalist parasitoids and hyperparasitoids tends to increase, possibly due to the spatial aggregation of mobile parasitoids toward areas of high host density (Eveleigh et al. 2007). In accordance with the *enemies' hypothesis*, it has been argued that this phenomenon is stronger in vegetationally diverse stands (Eveleigh et al. 2007), suggesting that complex host-parasitoids interactions may vary with forest diversity and/or forest fragmentation through its effects on parasitoid activity and dispersal.

Previous studies on spruce budworm parasitism have generally been undertaken at relatively local (i.e., stand) spatial scales (e.g., Miller 1963, Royama 1984, Eveleigh et al. 2007, Royama et al. 2017). As a result, we do not know if and how these relationships persist at broader spatial extents (e.g., the provincial scale; 1,000s of km²). A better understanding of the landscape-level interactions between forest vegetational diversity and spruce budworm parasitism is critical to developing a better understanding of how this system works, given that this is the scale at which spruce budworm outbreaks are manifest.

As a consequence of climate warming, spruce budworm activity is expected to increase at higher latitudes (Gray 2008, Régnière et al. 2012, Pureswaran et al. 2015), where both vegetational (Qian

and Ricklefs 2007) and lepidopteran host diversity (Kerr et al. 1998) are lower. In line with this prediction, most of the areas of severe defoliation in the current spruce budworm outbreak in eastern Canada are at higher latitudes than observed during the last century (Pureswaran et al. 2015). Reduced diversity of lepidopteran hosts for spruce budworm-associated parasitoids in northern forest landscapes could result in reduced spruce budworm parasitism relative to southern landscapes. The expected changes to spatial host-parasitoid interactions have unknown consequences on outbreak dynamics in these northern regions that have been little or unaffected by previous outbreaks (Pureswaran et al. 2015).

In this study, we investigated the associational effects of vegetational diversity on spruce budworm parasitism during the release phase of a spruce budworm outbreak in the province of Quebec, Canada. We specifically examined how spatial variation in forest diversity affects larval parasitism by two parasitoid wasp species that are generally thought to differ in their degree of host specialization. For each parasitoid species, we tested the *enemies' hypothesis* at the landscape level and developed regression models to describe how parasitism rates vary as a function of forest tree genus diversity, nonhost tree density, and spruce budworm outbreak age, all measured at multiple spatial scales. The results of our study will improve knowledge of the relative specialization of the two parasitoid species we examined, but more importantly, will improve our understanding of mechanisms behind spruce budworm outbreak dynamics, and will help us to better forecast outbreak consequences in novel northern habitats.

Materials and Methods

Study Organisms

The spruce budworm parasitoid community is very diverse (up to 66 primary parasitoids and 29 secondary and tertiary parasitoids [or hyperparasitoids]; Eveleigh et al. 2007). However, most parasitism of spruce budworm can be attributed to a few key species (Blais 1960, Eveleigh et al. 2014, Nealis 2016). This 'reduced' community can be partitioned in four temporal components according to the budworm stages attacked: eggs, early larvae, late larvae, or pupae (Blais 1960, Eveleigh and Johns 2014, Marrec et al. 2018). This study focuses on the early larval component, which is largely dominated by two wasp species during spruce budworm outbreaks: *Apanteles fumiferanae* (Viereck) (Hymenoptera: Braconidae) (Braconidae) and *Glypta fumiferanae* (Viereck) (Hymenoptera: Ichneumonidae) (Ichneumonidae) (Blais 1960, Eveleigh et al. 2014, Marrec et al. 2018). The high parasitism rates by *A. fumiferanae* (up to 56%) and *G. fumiferanae* (up to 24%) recorded in central Quebec during the last major outbreak in the 1970s over wide areas further suggest that in some environmental contexts, the combined action of these two species can exert an effective top-down control on spruce budworm larval populations (McLeod 1977).

Both species are univoltine with life cycles that match the spruce budworm (Brown 1946a,b). These species attack spruce budworm larvae in late summer and their progeny overwinter within second instar hosts (Lewis 1960, McLeod 1977). Immature parasitoids resume development as hosts become active the following spring (Lewis 1960) until they emerge from fourth to sixth instars (McGugan 1955), which is before most of other parasitoid species can attack the remaining hosts (Miller 1955, Eveleigh and Johns 2014). During their adult stage, the longevity and fecundity of both species increase when given access to food sources such as honey (Stairs 1983, Elliott et al. 1987, Nealis and Fraser 1988). This suggests that increased food sources in diverse forests could increase the

efficiency of both parasitoid species in regulating host populations (Russell 1989).

Despite having very similar life cycles and ecology, different studies have suggested that *A. fumiferanae* and *G. fumiferanae* may have different host ranges. Although both species have been described in the past as host-generalists (Krombein et al. 1979), Smith et al. (2011) recently demonstrated the presence of different cryptic species for *G. fumiferanae* that can be considered as host-specialists because each cryptic species attacks a different lepidopteran host species in the spruce budworm environment. Using the same approach, Smith et al. (2011) also showed that *A. fumiferanae* do not show cryptic diversity and that this species can exploit hosts from different lepidopteran families. Parasitism records from sampling spruce budworm larvae during endemic phases of spruce budworm population cycles further suggest a difference in host range between species.

During endemic periods, parasitism rates by *A. fumiferanae* are comparable to those obtained from outbreak records, while parasitism by *G. fumiferanae* is almost null (Miller and Renault 1976, Schaupp et al. 1991). This suggests that *A. fumiferanae* might be an effective *host switcher* and maintain large population sizes when spruce budworm populations are low, while *G. fumiferanae* does not appear to use this ecological strategy.

Insect Sampling

Early larval instars of the spruce budworm were sampled in 29 sites distributed within six different outbreak areas affected by defoliation in 2014 (Fig. 1a; Table 1). Timing of sampling was informed using the phenological model BioSIM (Régnière et al. 2013) such that spruce budworm larvae were sampled at the peak of the fourth larval instars, providing reliable parasitism rates for both species

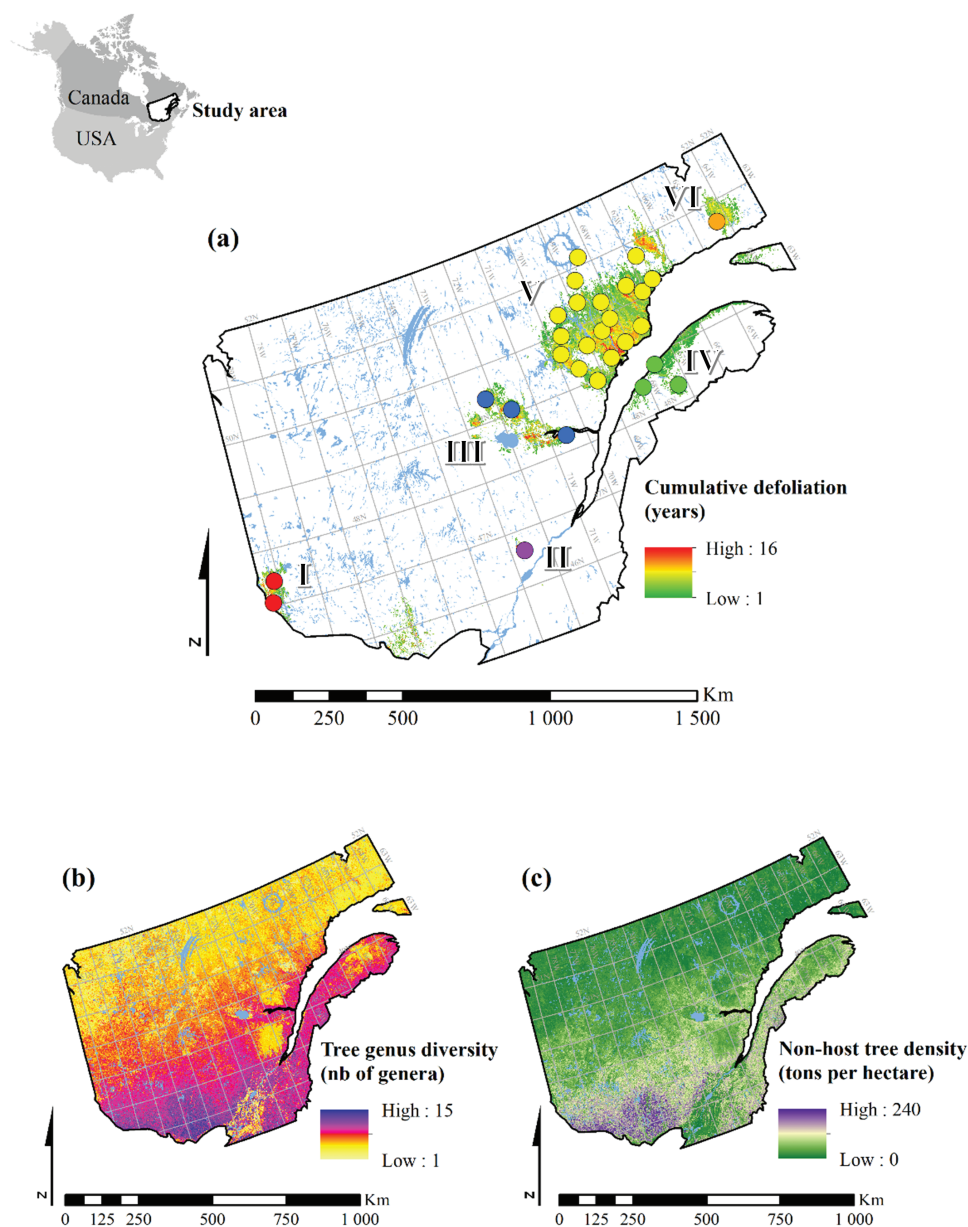


Fig. 1. (a) Study area in continental context showing sample sites across the province of Québec (Canada) and cumulative spruce budworm defoliation (see material and methods, spatial data at different spatial scales). Different colors of sites refer to different outbreak areas (see Table 1). (b and c) Maps of spatial variation in tree genus diversity (number of genus) and nonhost tree density (t/ha) over the extent of the study area. Data are from Beaudoin et al. (2014) and the Ministère des Forêts, de la Faune et des Parc (Color version of figure is available online).

Table 1. Sites coordinates, local spruce budworm larval density (mean number of larvae per bud; *SBWd*), number spruce budworm larvae reared (*Nsbw*), and parasitism rates by *Apanteles fumiferanae* and *Glypta fumiferanae*

Outbreak area	Site ID	Longitude	Latitude	SBWd	Nsbw	<i>A. fumiferanae</i>	<i>G. fumiferanae</i>
I. Abitibi-Témiscamingue	1	-79.11	47.42	0.17	339	0.14	0.01
	2	-79.27	47.05	0.20	343	0.12	0.01
II. Mauricie	3	-72.65	46.62	0.07	244	0.04	0.03
III. Saguenay-Lac-Saint-Jean	4	-72.23	49.40	0.25	168	0.05	0.00
	5	-71.66	49.06	0.27	257	0.05	0.07
	6	-70.54	48.28	0.11	252	0.12	0.03
IV. Bas-Saint-Laurent	7	-68.14	48.52	0.23	174	0.10	0.02
	8	-67.60	48.81	0.10	278	0.22	0.04
	9	-67.24	48.30	0.22	318	0.18	0.03
V. Côte-Nord	10	-69.19	48.96	0.31	345	0.07	0.08
	11	-69.53	49.28	0.30	361	0.07	0.06
	12	-69.84	49.65	0.47	396	0.06	0.14
	13	-69.64	49.95	0.23	351	0.05	0.12
	14	-67.49	49.54	0.75	367	0.13	0.14
	15	-68.60	49.24	0.32	283	0.03	0.07
	16	-69.08	49.61	0.43	333	0.02	0.06
	17	-68.07	49.39	0.17	242	0.07	0.09
	18	-68.54	49.75	0.22	223	0.04	0.04
	19	-68.20	49.90	0.41	286	0.04	0.04
	20	-68.24	50.23	0.42	259	0.04	0.11
	21	-69.49	50.32	0.27	290	0.02	0.03
	22	-68.85	50.39	0.11	396	0.01	0.04
	23	-68.66	50.77	0.07	82	0.04	0.07
	24	-68.31	51.14	0.03	19	0.00	0.05
	25	-67.05	50.09	0.34	288	0.01	0.14
	26	-67.40	50.31	0.28	185	0.01	0.05
	28	-66.66	50.22	0.37	244	0.01	0.18
	29	-66.80	50.72	0.50	211	0.03	0.17
VI. Basse-Côte-Nord	30	-64.30	50.62	0.19	144	0.00	0.16

of parasitoids (Miller 1955, Blais 1960). At each site, 10 dominant budworm host trees were randomly selected, and one 45-cm branch from the mid-crown of each tree was sampled to estimate local spruce budworm larval density. Sampling this strata has been shown to provide reliable estimates of spruce budworm densities (Morris 1955) and parasitism rates by *A. fumiferanae* and *G. fumiferanae* (Nealis and Régnière 1987, Eveleigh and Johns 2014). Additional larvae were sampled by taking extra branches (10–50, depending on local spruce budworm larval density) from mid-crown foliage. All branches were collected in paper bags and brought back to the laboratory for budworm identification and rearing.

Spruce budworm larvae were identified based on their morphological characteristics (Mackay 1962, Harvey and Stehr 1967) and expected developmental stages at the time of sampling (BioSIM; Régnière et al. 2013). We calculated the ratio of spruce budworm larvae to annual buds for each 45-cm branch. Ratios were then averaged for each site to provide an index of local spruce budworm larval density (SBWd). Spruce budworm larvae collected from 45-cm and extra branches were then reared individually on artificial diet. This product corresponds to Grisdale's modifications of the McMorran original recipe (Hervet et al. 2016). Based on the results from an experiment complementary to this study, we are confident that this rearing method did not affect our estimates of parasitism rates (see Supp Material 1). After egression from spruce budworm larvae, parasitoid cocoons were transferred to clean 1 oz. plastic cups until emergence of adult parasitoids. Parasitoid specimens were then identified to species (Brown 1946a,b, Mason 1974, Eveleigh et al. 2003, Bennett 2008, Fernández-Triana and Huber 2010). Several hyperparasitoid wasps of the genus *Stictopisthus* Thomson (Hymenoptera: Ichneumonidae) were retrieved from cocoons of

Apanteles sp. that emerged from diet-fed spruce budworm larvae. As these cocoons were identified as *A. fumiferanae* using the key provided by Eveleigh et al. (2003), these spruce budworm larvae were treated as parasitized by *A. fumiferanae* in further analyses. Parasitism rates were estimated for *A. fumiferanae* and *G. fumiferanae* at each sampling location by dividing the number of parasitized larvae by the number of reared larvae.

Voucher specimens have been deposited to the Ouellet-Robert entomological collection at the Université de Montréal.

Spatial Data at Different Spatial Scales

To describe the response of parasitoids to outbreak age, we used provincial monitoring datasets to measure cumulative defoliation at different spatial scales. Since 1968, systematic aerial surveys of spruce budworm defoliation have been made in the province of Québec by the Ministère de la Faune, des Forêts et des Parcs. These surveys are digitized and converted to polygon shapefiles (Ministère des Forêts, de la Faune et des Parcs 2016). Using these spatial data from 1993 to 2014, we rasterized (250 m) all polygons affected by moderate (35–69%) or severe (>69%) defoliation, and classified those pixels as defoliated. This resulted in 22 binary raster maps (one for each year; 1993–2014) that were then summed to calculate a cumulative defoliation map since the last major spruce budworm outbreak (CDEF; Fig. 1a).

To investigate relationships between forest diversity and parasitism, we used MODIS imagery (circa 2000–2006; 250 m; Beaudoin et al. 2014) to quantify forest structure at different spatial scales and to use as predictors in our models of parasitism rates. First, forest biodiversity was measured using information on 22 tree genera using raster layers of the percentage of aboveground tree biomass of the

22 tree genera present in the study area (*Abies*, *Acer*, *Alnus*, *Betula*, *Carpinus*, *Carya*, *Fagus*, *Fraxinus*, *Juglans*, *Larix*, *Ostrya*, *Picea*, *Pinus*, *Populus*, *Prunus*, *Quercus*, *Salix*, *Sorbus*, *Thuja*, *Tilia*, *Tsuga*, and *Ulmus*) were used to compute a metric of tree genus alpha diversity. For each genus-specific layer, pixel values were reclassified to represent genus presence or absence on the basis of an AGB > 1%. This threshold was selected to remove bias associated to rare genus. Layers were then combined and summed across the study area in a single map of pixel-level alpha diversity (TGD; Fig. 1b). Second, because the accuracy with which the different tree genera are measured varies (Beaudoin et al. 2014), we complemented the measure of forest alpha diversity with a layer representing nonhost tree (i.e., all genus except *Abies* and *Picea*) density expressed as tons per hectare (t/ha; Beaudoin et al. 2014; NHD; Fig. 1c).

Insect parasitoids perceive and respond to landscape composition at different spatial scales (Roland and Taylor 1997, Cronin and Reeve 2005). Thus, in order to identify relevant spatial scales for predicting parasitism rates spatial data on cumulative defoliation non-host tree density, and tree genus diversity values were summarized for each site using eight concentric circular buffers (1, 2, 3, 5, 7.5, 10, 15, and 20 km radius) centered on each sampling location (Fig. 2). We chose a maximum radius of 20 km because after this distance the buffers began to overlap with each other (Fig. 2). All these procedures were completed using ArcMap 10.3.1.

Statistical Analysis

The relative influences of latitude, longitude, and landscape factors (cumulative defoliation, tree genus diversity, and nonhost tree density) on parasitism by *A. fumiferanae* and *G. fumiferanae* were examined using a series of beta regressions as implemented in the *betareg* package (Cribari-Neto and Zeileis 2010) for R. This approach fits non-linear models using a beta distribution and logit link function, and is suitable to model parasitism rates as they are (0,1) bounded (Cribari-Neto and Zeileis 2010). All explanatory variables were standardized to zero mean and unit variance prior to analyses so that their relative effects on parasitism rates could be compared. Also, because sampling effort varied among sites, each site was weighted in each parasitism model by the number of spruce budworm larvae that were reared (Nsbw; Table 1). Finally, to test for an association between forest

diversity and SBWd, we used the same approach and set of predictors to model spruce budworm larval density (SBWd; mean ratio of spruce budworm larvae to annual buds). In this case, sites were weighted using the total number of buds counted in the 45-cm branches.

Models were built in two steps. In the first step, we identified the spatial scale at which each individual landscape factor was the most related to parasitism. For each species and each spatial scale, we independently modeled parasitism rates as a function of cumulative defoliation, tree genus diversity, or nonhost tree density using beta regression. We then retained scale-specific predictors associated with models with the higher weight of the Akaike's information criterion corrected for small sample sizes (AICc w) using the *AICcmodavg* package (Mazerolle 2017) in R. In the second step, we modeled spruce budworm larval density or parasitism rates as a function of different combinations of both scale-specific predictors filtered in the first step, and the other predictors that were independent of spatial scale (i.e., latitude, longitude, and SBWd for parasitism rates). When factors were highly correlated (i.e., Pearson's correlation ≥ 0.70), only one factor was retained in the final list of candidate models. The final lists, each including a null model, are listed in Table 3. We retained for each species the model with the lower $\Delta AICc$, and the higher AICc w and pseudo R^2 . This last metric is a global measure of explained variation (Cribari-Neto and Zeileis 2010).

Finally, to validate the premise that budworm outbreaks do indeed vary with forest diversity (Cappuccino et al. 1998, Quayle et al. 2003, Bouchard and Auger 2014, Robert et al. 2018) in our study area, we repeated the modeling procedures described above using spruce budworm larval density as the response variable. In this case, sites were weighted using the total number of buds counted in the 45-cm branches.

Results

Spruce Budworm Larval Density

Spruce budworm larval densities varied considerably between sampling sites. Densities varied from 0.03 to 0.75 larvae per bud (Table 1), with a mean value of 0.27 (SD = 0.15). Spruce budworm densities were generally higher in outbreak areas V (Table 1; Fig. 1a). As expected, spruce budworm larval densities were lower in more

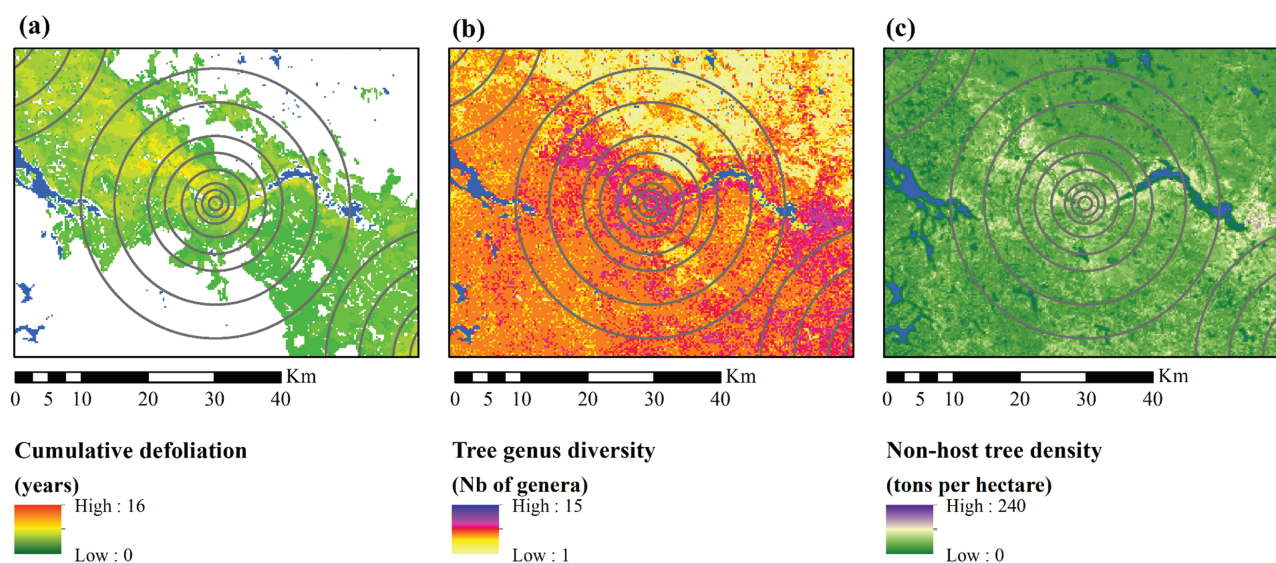


Fig. 2. Maps showing buffer areas around site no. 11 used to characterize the mean value of pixels (250 m) for (a) cumulative defoliation (number of years), (b) tree genus diversity (number of genus), and (c) nonhost tree density (t/ha) (Color version of figure is available online).

diverse forests. The best model explaining spruce budworm larval density included cumulative defoliation at a scale of 20 km and non-host tree density also at 20 km (AICc $w = 1.00$, pseudo $R^2 = 0.19$). The predictor with the highest coefficient was cumulative defoliation, with a significant positive effect on spruce budworm larval density. Nonhost tree density was also a significant predictor, with a negative effect on larval density (Supp Material 2 online only).

Parasitism

We successfully reared 7,691 spruce budworm larvae on artificial diet. From these, 7.42% were parasitized by *A. fumiferanae* and 7.35% by *G. fumiferanae*. Of the *A. fumiferanae* cocoons emerged from spruce budworm larvae, 3.87% were parasitized by *Stictopisthus* sp. (29/750). No hyperparasitoids were collected from *G. fumiferanae*. Finally, only 0.55% of spruce budworm larvae were parasitized by other species of parasitoids or unidentifiable immature parasitoids. Other species of parasitoids included, in decreasing order of importance, *Tranosema tenuifemur* (Walley) (Hymenoptera: Ichneumonidae), *Smidtia fumiferanae* (Tohill) (Diptera: Tachinidae), *Sacrophaga aldrichi* (Parker) (Diptera: Sacrophagidae), and the braconids *Apanteles petrovae* (Walley) (Hymenoptera: Braconidae), *Bassus acrobasis* (Cushman) (Hymenoptera: Braconidae) and *Bassus binominatus* (Muesebeck) (Hymenoptera: Braconidae).

Parasitism rates by *A. fumiferanae* ranged from 0 to 22% (Table 1), with a mean value of 6.10% (SD = 5.51%). Mean parasitism rates were higher in the southernmost outbreak areas, particularly in area IV, which is the most recently defoliated (Table 1; Fig. 1a). Parasitism was almost null at the northern margins of the outbreak area V, where defoliation is also recent, and in outbreak area VI (Table 1; Fig. 1a).

Parasitism rates by *G. fumiferanae* ranged from 0 to 18% (Table 1), with a mean value of 7.17% (SD = 5.18%). In contrast with *A. fumiferanae*, parasitism rates were higher in outbreak areas V and VI, and lower in the southernmost outbreak areas I, II, III, and IV (Table 1; Fig. 1a).

We found no significant correlation between rate of parasitism by *A. fumiferanae* and *G. fumiferanae* (Weighted Pearson's correlation: -0.24 ; $t = -1.27$; $P = 0.2133$; Fig. 3).

Scale-Specific Analyses

During the first modeling step using beta regression and AICc w , we found that cumulative defoliation influences parasitism rates by both *A. fumiferanae* and *G. fumiferanae* at a scale of 20 km (Fig. 4a and d). At this scale, cumulative defoliation had a significant negative effect on parasitism by *A. fumiferanae* and a significant positive effect on parasitism by *G. fumiferanae* (results not shown). Using the same procedure, tree genus diversity and nonhost tree density metrics were selected for both parasitoid species. Parasitism rates by *A. fumiferanae* were best explained by tree genus diversity at a scale of 3 km (Fig. 4b) or nonhost tree density at 15 km (Fig. 4c; positive effects). In contrast, parasitism rates by *G. fumiferanae* were best explained by tree genus diversity at 20 km (Fig. 4e) or nonhost tree density at 15 km (Fig. 4f; negative effects).

Model Comparisons

We found significant correlations between spatial predictors retained at step 1 and predictors independent of spatial scale. Metrics describing tree genus diversity, nonhost tree density, latitude, and longitude were all correlated (Table 2). Therefore, we did not combine those factors in the final list of candidate models (Table 3). Also, because latitude and longitude associated with our sampling locations were correlated (0.7), we chose not to include longitude in the analyses

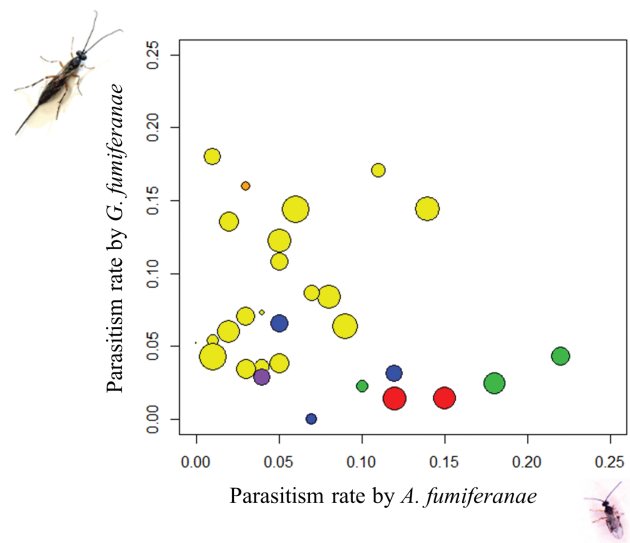


Fig. 3. Relationship between parasitism rates by *Apanteles fumiferanae* and *Glypta fumiferanae* (weighted Pearson's correlation: -0.24 ; $t = -1.27$; $P = 0.2133$). Color of points corresponds to the different outbreak areas (Fig. 1), and size of points is proportional to the number of spruce budworm larvae that were reared to estimate parasitism rates (Nsbw; Table 1) (Color version of figure is available online).

and focussed on latitude as this factor best fits our research questions regarding the effects of latitudinal gradients of vegetational diversity on parasitism rates of the spruce budworm.

Based on AICc w , the best model explaining parasitism by *A. fumiferanae* included tree genus diversity measured at 3 km, spruce budworm larval density, and cumulative defoliation measured at 20 km (AICc $w = 1.00$, pseudo $R^2 = 0.59$; Table 3). For this model, the predictor with the highest coefficient was tree genus diversity, with a significant positive effect on parasitism rates. The coefficient associated with spruce budworm larval density was also positive and significant. Cumulative defoliation had a significant negative effect on parasitism rates (Table 4; Fig. 5).

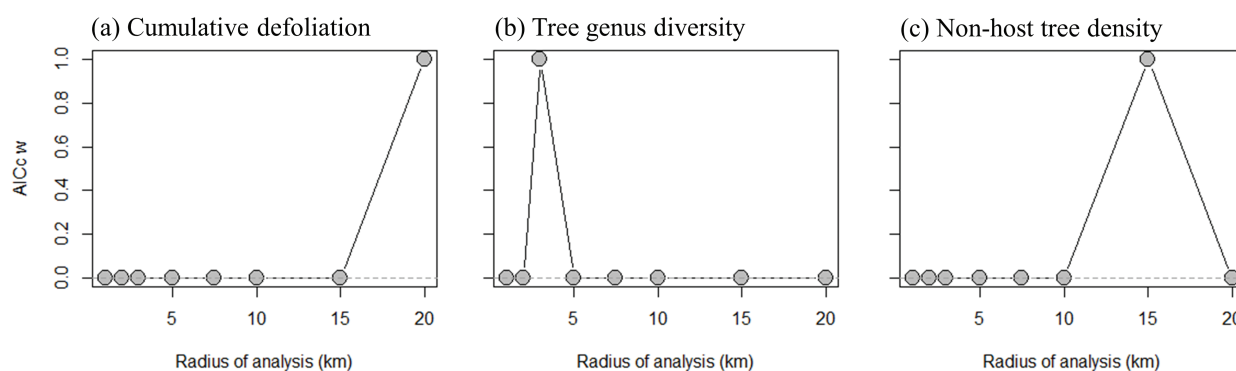
For *G. fumiferanae*, the best model included nonhost tree density measured at a scale of 20 km, spruce budworm larval density, and cumulative spruce budworm defoliation measured at 20 km as predictors (AICc $w = 0.99$, adjusted $R^2 = 0.51$; Table 4). For this species, nonhost tree density had a significant negative effect on parasitism rates, while SBWd had a significant positive effect. Cumulative defoliation had only a marginal, but significant positive effect on parasitism rates (Table 4; Fig. 5).

Discussion

In this study, we tested the *enemies' hypothesis* (Root 1973) that vegetational diversity is positively related to top-down control on spruce budworm populations through an indirect effect on natural enemies. To do so, we modeled spatial variation of parasitism rates by *A. fumiferanae* and *G. fumiferanae* as a function of forest diversity. We found that parasitism by *A. fumiferanae* and *G. fumiferanae* respond in opposite ways to this gradient in forest diversity. Parasitism by *A. fumiferanae* was higher in the south where both vegetational (Qian and Ricklefs 2007) and lepidopteran (Kerr et al. 1998) diversities are known to be higher, whereas the opposite pattern was observed for *G. fumiferanae*. Differences in parasitism rates as a function of forest diversity could have important influences on the spruce budworm-parasitoid food-web structure and outbreak dynamics (Eveleigh et al. 2007).



Parasitism by *Apanteles fumiferanae*



Parasitism by *Glypta fumiferanae*

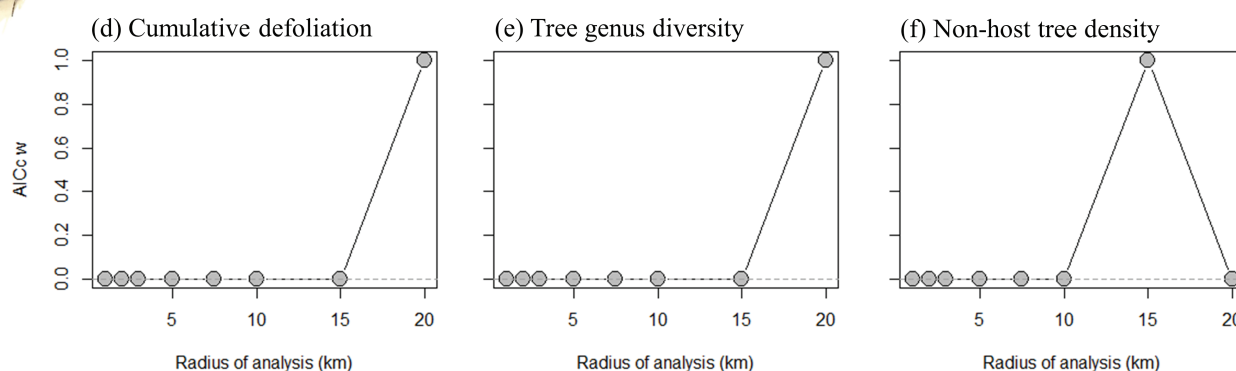


Fig. 4. Plots showing $AICc$ weights of preliminary beta regressions models of parasitism rates by *Apanteles fumiferanae* (a–c) and *Glypta fumiferanae* (d–f) as a function of landscape-level factors measured at different spatial scales. In each plot, the gray dashed line shows the value of the $AICc$ weight of the intercept model.

Table 2. Pairwise Pearson's correlation coefficients between predictor variables selected during the first step of modeling for *Apanteles fumiferanae* and *Glypta fumiferanae*

<i>A. fumiferanae</i>	LAT	LONG	TGD ₃	NHD ₁₅	CDEF ₂₀	SBWd
LAT	-					
LONG	0.70	-				
TGD ₃	-0.90	-0.59	-			
NHD ₁₅	-0.94	-0.65	0.94	-		
CDEF ₂₀	0.06	0.03	-0.06	-0.08	-	
SBWd	0.22	0.25	-0.19	-0.28	0.34	-
<i>G. fumiferanae</i>	LAT	LONG	TGD ₂₀	NHD ₁₅	CDEF ₂₀	SBWd
LAT	-					
LONG	0.70	-				
TGD ₂₀	-0.92	-0.53	-			
NHD ₁₅	-0.94	-0.65	0.94	-		
CDEF ₂₀	0.06	0.03	-0.19	-0.08	-	
SBWd	0.22	0.25	-0.31	0.28	0.34	-

To minimize multi-collinearity in our models during the second step, we did not include pairs of variables that exhibited correlations ≥ 0.70 (bolded).

LAT = latitude; LONG = longitude; TGD = tree genus diversity; NHD = nonhost tree density; SBWd = spruce budworm larval density; CDEF = cumulative defoliation (for the latter tree factors, the spatial scale in km selected during the first step of modeling is indicated by a subscript).

Spruce Budworm Larval Density

We found that cumulative spruce budworm defoliation surrounding the sampling site was the best predictor explaining local spruce budworm larval density, followed by nonhost tree

density ([Supp Material 2 online only](#)). This result coincides with the recent scientific literature concerning the release of spruce budworm outbreaks. In their spatial analysis in the province of Quebec between 2003 and 2011, [Bouchard and Auger \(2014\)](#)

Table 3. List of candidate models and their relative support describing variation in rates of spruce budworm parasitism by *Apanteles fumiferanae* and *Glypta fumiferanae*

Species	Model	Parameters	Pseudo R^2	AICc	Δ AICc	AICc w
<i>A. fumiferanae</i>	1	TGD ₃ + SBWd + CDEF ₂₀	0.59	-32,258	0.00	1.00
	2	TGD ₃ + SBWd	0.58	-31,949	308.94	0.00
	3	TGD ₃ + CDEF ₂₀	0.52	-31,259	999.53	0.00
	4	TGD ₃	0.52	-31,238	1,019.80	0.00
	5	NHD ₁₅ + SBWd + CDEF ₂₀	0.51	-30,283	1,974.73	0.00
	6	NHD ₁₅ + SBWd	0.50	-30,100	2,158.26	0.00
	7	NHD ₁₅	0.41	-29,136	3,121.79	0.00
	8	NHD ₁₅ + CDEF ₂₀	0.41	-29,134	3,124.42	0.00
	9	LAT + SBWd + CDEF ₂₀	0.42	-28,766	3,492.53	0.00
	10	LAT + SBWd	0.41	-28,646	3,612.43	0.00
	11	LAT + CDEF ₂₀	0.35	-28,110	4,148.05	0.00
	12	LAT	0.35	-28,110	4,148.31	0.00
	13	SBWd + CDEF ₂₀	0.01	-25,810	6,448.34	0.00
	14	CDEF ₂₀	0.00	-25,769	6,488.65	0.00
	15	SBWd	0.00	-25,663	6,594.88	0.00
	16	Intercept model	NA	-25,661	6,596.63	0.00
<i>G. fumiferanae</i>	1	NHD ₁₅ + SBWd + CDEF ₂₀	0.51	-32,551	0.00	0.99
	2	NHD ₁₅ + SBWd	0.51	-32,541	9.94	0.01
	3	LAT + SBWd + CDEF ₂₀	0.46	-31,827	724.04	0.00
	4	LAT + SBWd	0.46	-31,817	734.21	0.00
	5	TGD ₂₀ + SBWd + CDEF ₂₀	0.43	-31,426	1,125.85	0.00
	6	TGD ₂₀ + SBWd	0.43	-31,425	1,126.46	0.00
	7	NHD ₁₅ + CDEF ₂₀	0.41	-30,567	1,984.56	0.00
	8	NHD ₁₅	0.39	-30,285	2,266.52	0.00
	9	SBWd + CDEF ₂₀	0.25	-29,639	2,912.09	0.00
	10	SBWd	0.25	-29,636	2,915.02	0.00
	11	LAT + CDEF ₂₀	0.33	-29,602	2,949.28	0.00
	12	TGD ₂₀ + CDEF ₂₀	0.33	-29,312	3,239.27	0.00
	13	LAT	0.31	-29,272	3,279.24	0.00
	14	TGD ₂₀	0.32	-29,172	3,379.28	0.00
	15	CDEF ₂₀	0.04	-26,827	5,724.13	0.00
	16	Intercept model	NA	-26,376	6,175.13	0.00

For each species, models are ordered according to the weight of Akaike's information criterion corrected for small sample sizes (AICc w).

LAT = latitude; LONG = longitude; TGD = tree genus diversity; NHD = nonhost tree density; SBWd = spruce budworm larval density; CDEF = cumulative defoliation (for the latter tree factors, the spatial scale in km selected during the first step of modeling is indicated by a subscript).

Table 4. Beta regression parameter estimations ($\pm 95\%$ confidence intervals) for retained models describing parasitism rates of spruce budworm larvae by *Apanteles fumiferanae* and *Glypta fumiferanae*

Species	Predictor	Estimate	95% CI	Z value	Pr(> z)
<i>A. fumiferanae</i>	TGD ₃	0.4841	0.4741; 0.4940	95.42	<0.001
	SBWd	0.1719	0.1627; 0.1811	36.65	<0.001
	CDEF ₂₀	-0.1046	-0.1155; -0.0936	-18.70	<0.001
<i>G. fumiferanae</i>	TGD ₂₀	-0.3393	-0.3393; -0.3177	-59.70	<0.001
	SBWd	0.2842	0.2754; 0.2930	63.31	<0.001
	CDEF ₂₀	0.0220	0.0115; 0.0325	4.12	<0.001

demonstrated that the spread of spruce budworm defoliation in landscapes is favored by adjacent defoliation and restrained by the proportion of hardwood (or nonhost) tree species. This phenomenon can be explained by a lower probability that dispersing moths from surrounding areas lay eggs on appropriate host trees (Nealis and Régnière 2004), or, as discussed in this study and others (Cappuccino et al. 1998, Quayle et al. 2003, Eveleigh et al. 2007), by higher top-down control of spruce budworm populations by natural enemies in vegetationally diverse forests (i.e., the *enemies' hypothesis*).

Host Specificity and Parasitism Rates of Spruce Budworm Larvae

We found that tree genus diversity was the best predictor of parasitism by *A. fumiferanae*, with a positive effect (Table 4; Fig. 5a). Host-generalism may explain why vegetational diversity had such a strong positive effect on parasitism rates. *A. fumiferanae* is known to attack multiple host species from different lepidopteran families (Mason 1974, Krombein et al. 1979, Smith et al. 2011). Additionally, many of *A. fumiferanae*'s hosts are generalist defoliators (Robinson et al. 2010). Because southern forest landscapes sustain a greater

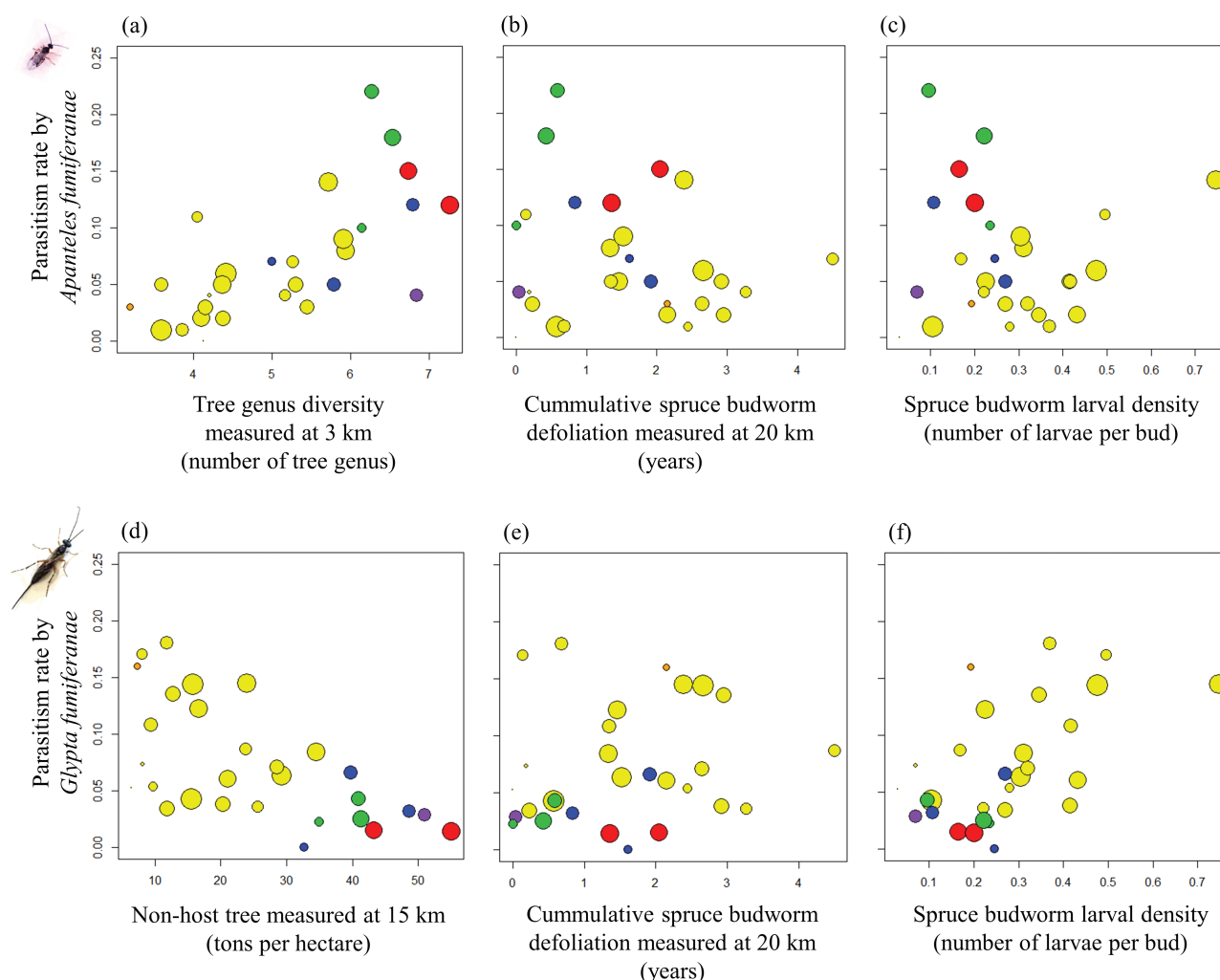


Fig. 5. Relationships between parasitism rates by *Apanteles fumiferanae* (a–c) and *Glypta fumiferanae* (d–f) as a function of parameters identified in the optimal beta regression models. Color of points corresponds to the different outbreak areas (Fig. 1), and size of points is proportional to the number of spruce budworm larvae that were reared to estimate parasitism rates (*Nsbw*; Table 1) (Color version of figure is available online).

diversity of Lepidoptera (Kerr et al. 1998), whose larvae can act as alternative hosts for parasitoids, it is reasonable to expect a positive association between forest diversity and parasitism rates for *A. fumiferanae*. More importantly, host-generalism could allow this species to remain at relatively high population densities in diversified landscapes during endemic periods of spruce budworm population cycles (Miller and Renault 1976) and therefore inflict high parasitism rates when outbreaks are still young (e.g., in outbreak area IV; Fig. 1; Table 1).

Parasitism by *A. fumiferanae* was also affected by spruce budworm-related factors. Spruce budworm larval density had a positive effect on parasitism rates, while outbreak age (or cumulative defoliation) had a negative effect when measured at a radius of 20 km. These results suggest that the density dependence of *A. fumiferanae* on SBWd may decrease with time during an outbreak, possibly due to the effects of increasing competition with other parasitoids species and/or increasing mortality due to generalist hyperparasitoids (Eveleigh et al. 1994, Eveleigh et al. 2007). Together, these results suggest that *A. fumiferanae* could remain a relatively minor mortality factor of spruce budworm larvae in northern outbreaks in the future.

Parasitism rates by *G. fumiferanae* were best predicted by SBWd (positive effect) and nonhost tree density (negative effect; Table 4;

Fig. 5). This pattern could be explained by the host- or habitat-specificity of *G. fumiferanae*. A recent study has demonstrated the presence of different cryptic species of *G. fumiferanae* that are host-specialists and attack different lepidopteran host species in the spruce budworm environment (Smith et al. 2011). A potential explanation to this phenomenon could be that genetic variation within the species complex might be associated with different foraging behaviors of female parasitoids that allow them to persist in lower diversity coniferous stands that are more susceptible to the spruce budworm (Bouchard and Auger 2014). As a result, host specificity of *G. fumiferanae* cryptic species could explain the stronger host density dependence of *G. fumiferanae* parasitism rates (Table 4) and the near absence of this species from parasitoid rearings during endemic phases of spruce budworm population cycles (Miller and Renault 1976).

An important point worthy of discussion here is that, since our study area is large, the variables associated with forest diversity and composition reflect a latitudinal gradient (Table 2). It is therefore possible that the relationships we identified between parasitism and forest diversity mask other phenomena associated with effects of temperature (Nyrop and Simmons 1986, Nealis 1988) or precipitation regimes (Stireman et al. 2005) on parasitoid activity. However,

our regression models of parasitism based solely on latitude performed less well than those based on forest diversity. We are therefore confident that we have effectively tested the *enemies' hypothesis* for both parasitoid species.

Spatial Scale and Parasitoid Dispersal

The different spatial scales at which forest composition affects parasitism could be related to differences in dispersal capacities of parasitoid species, as has been shown for the dipteran parasitoids of the forest tent caterpillar, *Malacosoma disstria* (Lepidoptera: Lasiocampidae) (Roland and Taylor 1997). In this parasitoid complex, the scale at which parasitism most strongly responds to forest structure corresponds to the relative body size of fly species (Roland and Taylor 1997). Another well-studied host–parasitoid system support this hypothesis: the small parasitoid *Cotesia melitaearum* (Hymenoptera: Braconidae) and the larger *Hyposoter horticola* (Hymenoptera: Ichneumonidae) both attack the endangered Glanville fritillary butterfly, *Melitaea cinxia* (Lepidoptera: Nymphalidae) in the Åland Islands of Finland. In this region, the smaller braconid has a lower dispersal ability (<1 km/yr) than the larger ichneumonid (>5 km/yr; (Lei and Hanski 1998).

In line with the results of the previous studies, the scale at which forest diversity best described rates of parasitism differed between *A. fumiferanae* and *G. fumiferanae* (Fig. 4). Parasitism rates were best explained by tree genus diversity at a 3 km radius for *A. fumiferanae* and by nonhost tree density at a radius of 15 km for *G. fumiferanae* (Fig. 4b and f). The smaller *A. fumiferanae* (~3.5 mm in length; Brown 1946a) could have lower dispersal capacities than the larger *G. fumiferanae* (~8.0 mm in length; Brown 1946b). Parasitoids of the spruce budworm might disperse from defoliation epicenters toward low defoliation areas. Reduced relative dispersal in the smaller *A. fumiferanae*, combined with resistance associated with forest diversity, could explain why this species causes lower parasitism rates in the northernmost sites relative to *G. fumiferanae*, for which parasitism rates increase in landscapes with low vegetational diversity. However, this analysis requires cautious interpretation as the average values of tree genus diversity or nonhost tree density did not differ strongly among analysis radii (not shown). Thus, it would be interesting to estimate parasitoid dispersal capabilities in landscapes using alternative methods, such as marking and tracking (Lavandero et al. 2004), or molecular markers (Kankare et al. 2005).

A Competition-Dispersal Trade-Off?

All species are to some extent limited by their resources or their natural enemies (Amarasekare 2003). Species coexistence in communities can be achieved by interspecific trade-off between species traits, such as dispersal and competition (Kneitel and Chase 2004). A competition-dispersal trade-off in *A. fumiferanae* and *G. fumiferanae* could explain the coexistence of both species in landscapes when spruce budworm population are low and hosts are more limiting. According to this hypothesis, *A. fumiferanae* would be expected to be the stronger intrinsic competitor and *G. fumiferanae* the better disperser. A previous study has indeed shown that *A. fumiferanae* larvae can neutralize the development of the immature stages of *G. fumiferanae* when both species are parasitizing the same host larvae (Lewis 1960). In contrast, the larger body size of *G. fumiferanae* (Brown 1946a,b) may allow this species to avoid competition with *A. fumiferanae* by colonizing forested landscapes more recently affected by the spruce budworm.

Although limited, field studies on spruce budworm parasitism during endemic stages also seem to support a competition-dispersal trade-off between *A. fumiferanae* and *G. fumiferanae*. When spruce

budworm population densities are very low, parasitism by *A. fumiferanae* in the province of New Brunswick can often reach 20%, while parasitism by *G. fumiferanae* is almost never higher than 1% (Miller and Renault 1976).

In the context of our study, however, host resources are not limiting for *A. fumiferanae* nor *G. fumiferanae*, as maximum combined percentage of parasitism was only 27.0% (Site no. 14; Table 1), and correlation between parasitism rates was not significant (Fig. 3). Also, McLeod (1977) found that the frequency of multi-parasitism involving both species was very low during the peak of the last outbreak during the 1970s, which is a surprising result considering that parasitism rates by both parasitoid species were much higher than in our study. Thus, interspecific competition alone cannot explain the contrasting spatial patterns in parasitism rates by *A. fumiferanae* and *G. fumiferanae* that we presented in this study.

Apparent Competition Through Hyperparasitism?

Another factor that could explain lower parasitism by *G. fumiferanae* in more diverse landscape is hyperparasitism. Although we reported hyperparasitism only for *A. fumiferanae*, it is important to note that hyperparasitism associated with *A. fumiferanae* or *G. fumiferanae* occur mainly during the cocoon stage (S. Legault, unpublished data). Heterogeneous stands have been shown to sustain a greater diversity of hyperparasitoids during spruce budworm outbreaks, with many of them attacking both *A. fumiferanae* and *G. fumiferanae* (Eveleigh et al. 2007, Smith et al. 2011). However, mortality rates of *G. fumiferanae* cocoons could be higher than for *A. fumiferanae* since the cocoon of the former species is much thinner (Brown 1946a,b) and thus could be less resistant to oviposition by hyperparasitoids.

Conclusions

With global warming, spruce budworm outbreaks are expected to shift toward higher latitudes (Gray 2008, Régnière et al. 2012, Pureswaran et al. 2015, Boulanger et al. 2016). As a result, spruce budworm outbreaks will be found in regions that are generally less floristically diverse than within the traditional outbreak range. Through analysis of spatial variation in parasitism by two principal spruce budworm parasitoids, we identified a contrasting effect of forest diversity in parasitism rates by a presumed high-disperser host-specialist (*G. fumiferanae*) versus a weak-disperser generalist parasitoid (*A. fumiferanae*). Tree genus diversity had a positive influence on *A. fumiferanae*, while nonhost tree density had a negative effect on *G. fumiferanae*. Although *A. fumiferanae* is the dominant parasitoid species in the central part of the spruce budworm outbreak range (Table 1; Fig. 1; McLeod 1977), we demonstrated that this species is progressively replaced by *G. fumiferanae* in northern landscapes. However, this study focussed only on the early larval component of the spruce budworm-parasitoids community. It would be very interesting in the future to test the *enemies' hypothesis* for other common parasitoid species (for example *Dirophanes maculicornis* (Stephens) (Hymenoptera: Ichneumonidae) vs. *Smidtia fumiferanae* (Tothill) (Diptera: Tachinidae; Marrec et al. 2018) by sampling later stages of the spruce budworm throughout its outbreak range.

Nevertheless, our results show that a forest diversity gradient in *A. fumiferanae* versus *G. fumiferanae* dominance could lead to more severe annual defoliation during future outbreaks as the budworm's range shifts northward. Indeed, *A. fumiferanae* kills its hosts earlier in the development of the spruce budworm (larval instars III to V) than *G. fumiferanae* (instars V and VI; McGugan 1955). This

difference means that *A. fumiferanae* is more effective preventing annual defoliation of tree foliage than *G. fumiferanae*, because most of spruce budworm feeding occur during the sixth larval instar (Albert and Baucé 1994). To maximize parasitism rates of *A. fumiferanae* during future outbreaks, we suggest considering the *enemies' hypothesis* for the management of public forests in northern landscapes by maximizing forest diversity at spatial scales relevant to parasitoids' dispersal abilities. Increasing tree diversity is also a promising approach to increase the resilience of forests to other natural disturbances such as droughts, forest fires, windstorms, pathogens, and invasive species (Jactel et al. 2017, Astrup et al. 2018). Such changes could occur naturally through reduced fire suppression in the boreal forest or through the establishment of mixed stands through artificial regeneration (Astrup et al. 2018).

Supplementary Data

Supplementary data are available at *Environmental Entomology* online.

Acknowledgments

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